Types of Errors in Inferences and Power of a Statistical Test

I. The Idea

II. Two Types of Errors

III. What Determines a Test's Power
   A. Size of Type I Error
   B. Size of Treatment effect
   C. Population Variability
   D. Number of Samples

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  - The chi-squared test
    - goodnessOfFitTest()
    - associationTest()
    - **chisq.test()**
- Homework 2 Overview and Guidance

I. The Idea

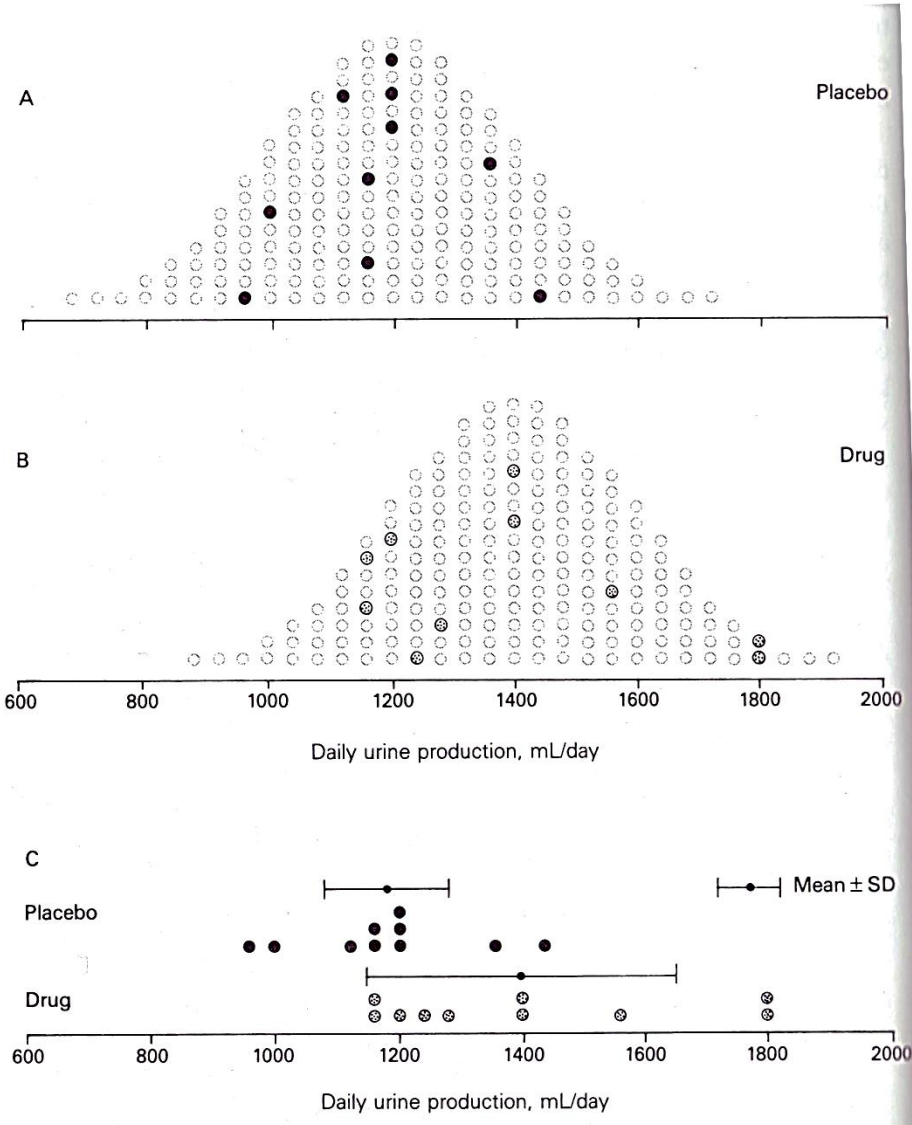


Figure 6-1 Daily urine production in a population of 200 people while they are taking a placebo and while they are taking an effective diuretic that increases urine production by 200 mL/day on the average. Panels *A* and *B* show the specific individuals selected at random for study. Panel *C* shows the results as they would appear to the investigator. $t = 2.447$ for these observations. Since the critical value of t for $P < .05$ with $2(10 - 1) = 18$ degrees of freedom is 2.101, the investigator would probably report that the diuretic was effective.

$$\mu_1 = 1200 \text{ mL/day}$$

$$\sigma_1 = 200 \text{ mL/day}$$

$$\mu_2 = 1400 \text{ mL/day}$$

$$\sigma_2 = 200 \text{ mL/day}$$

$$\bar{x}_1 = 1180 \quad s_1 = 144 \quad n_1 = 10$$

$$\bar{x}_2 = 1400 \quad s_2 = 245 \quad n_2 = 10$$

$$|t| = 2.447 \quad \nu = 18 \quad t_{0.05}(18) = 2.101$$

Reject Null Hypothesis - Correct

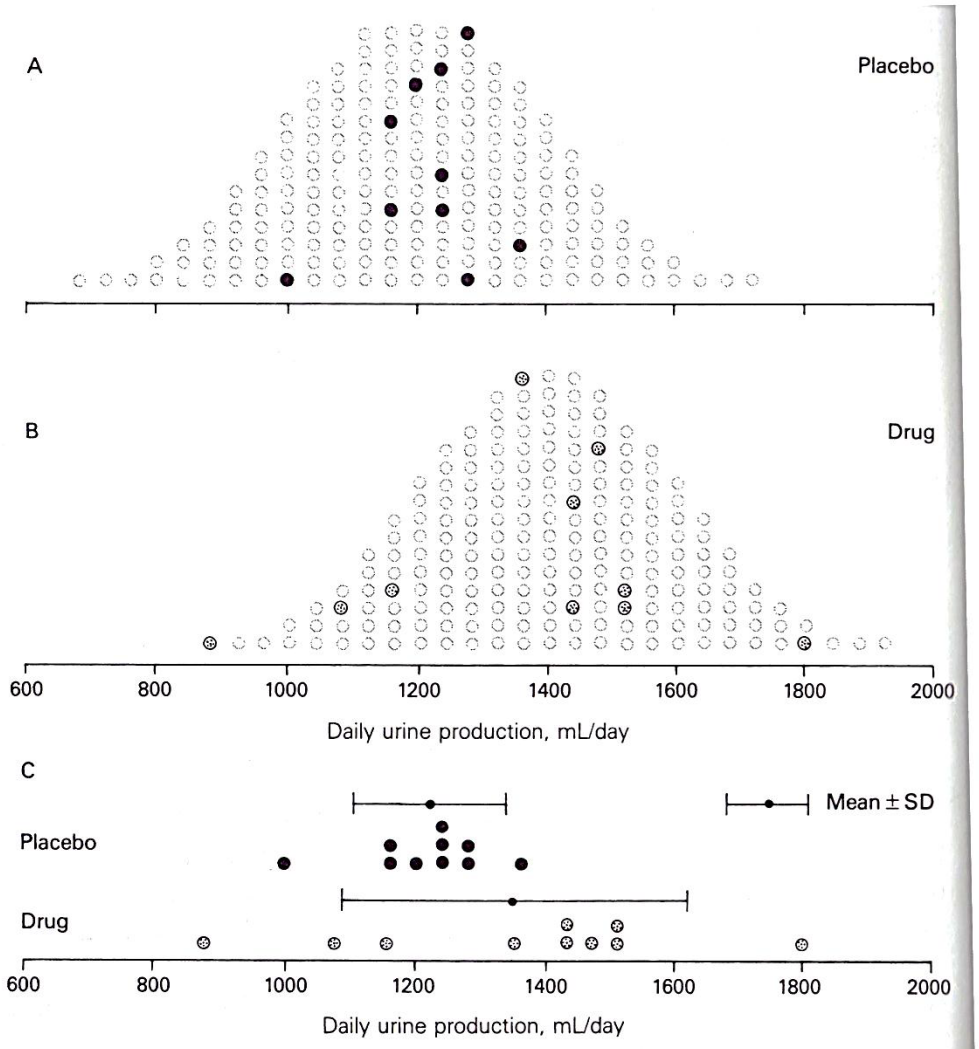


Figure 6-2 There is nothing special about the two random samples shown in Fig. 6-1. This illustration shows another random sample of two groups of 10 people each selected at random to test the diuretic and the results as they would appear to the investigator. The value of t associated with these observations is only 1.71, not great enough to reject the hypothesis of no drug effect with $P < 0.05$, that is, $\alpha = 0.05$. If the investigator reported the drug had no effect, he/she would be wrong.

$$\mu_1 = 1200 \text{ mL/day}$$

$$\sigma_1 = 200 \text{ mL/day}$$

$$\mu_2 = 1400 \text{ mL/day}$$

$$\sigma_2 = 200 \text{ mL/day}$$

$$\bar{x}_1 = 1216 \quad s_1 = 97 \quad n_1 = 10$$

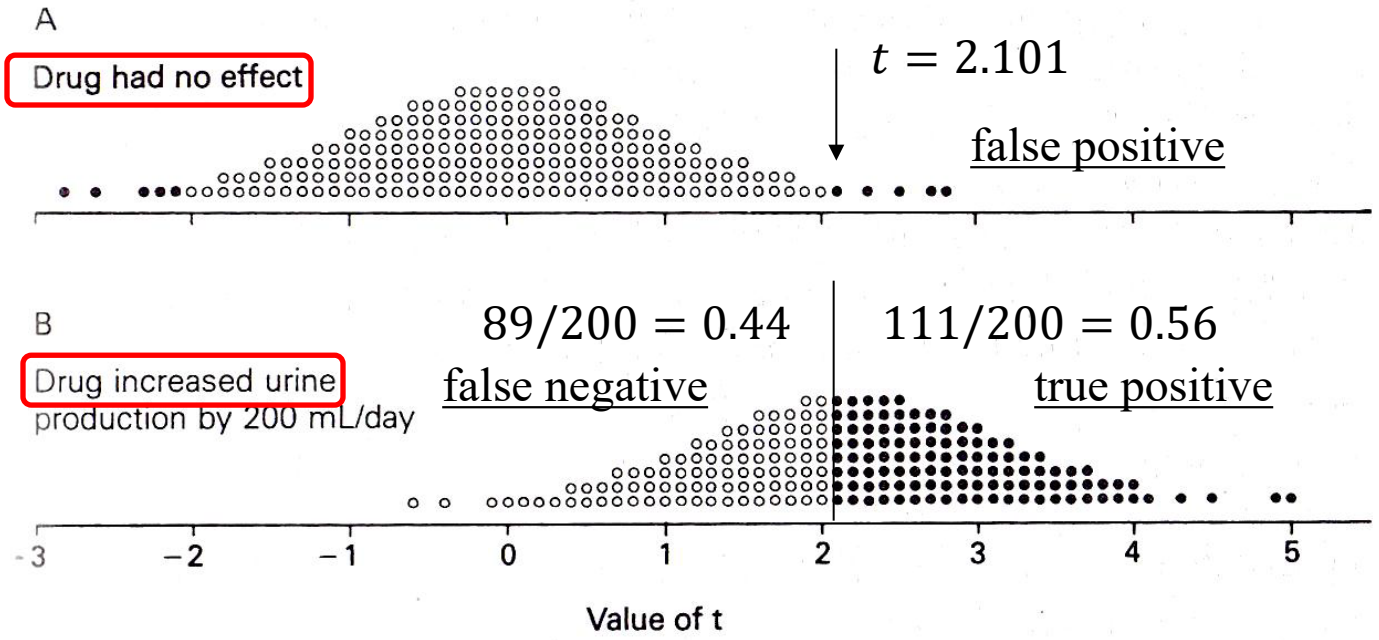
$$\bar{x}_2 = 1368 \quad s_2 = 263 \quad n_2 = 10$$

$$|t| = 1.71 \quad \nu = 18 \quad t_{0.05}(18) = 2.101$$

Do not reject Null Hypothesis - <u>Incorrect</u>

Given a method that produces < 0.05 chance of false positive what is the chance of a false negative?

Distribution of *t* Values



$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s^2}{n_1} + \frac{s^2}{n_2}}}$$

$$s^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

$$\nu = n_1 + n_2 - 2$$

Figure 6-3 (*A*) The distribution of values of the *t* statistic computed from 200 experiments that consisted of drawing two samples of size 10 each from a single population; this is the distribution we would expect if the diuretic had no effect on urine production. (Compare with Fig. 4-5*A*.) (*B*) The distribution of *t* values from 200 experiments in which the drug increased average urine production by 200 mL/day. $t = 2.1$ defines the most extreme 5 percent of the possible values of *t* when the drug has no effect; 111 of the 200 values of *t* we would expect to observe from our data fall above this point when the drug increases urine production by 200 mL/day. Therefore, there is a 55 percent chance that we will conclude that the drug actually increases urine production from our experiment.

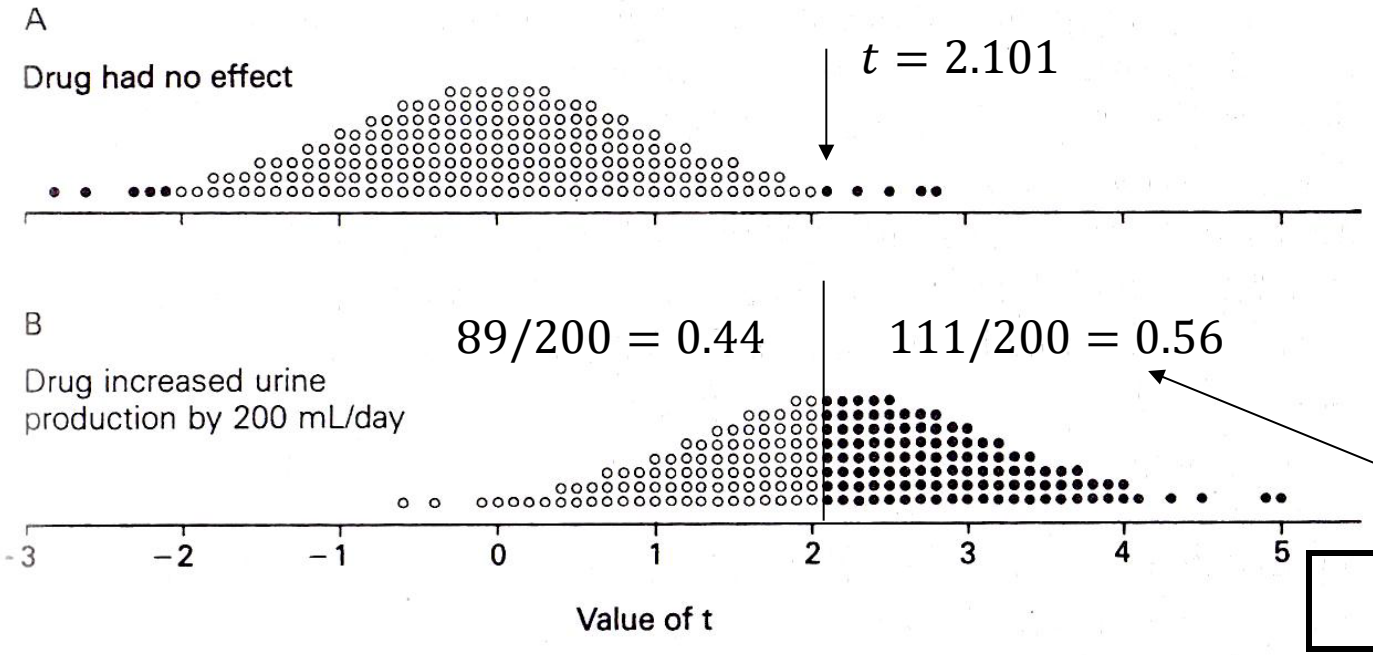
II. Two Types of Errors

Table 6-1 Types of Erroneous Conclusions in Statistical Hypothesis Testing

	Actual situation	
Conclude from observations	Treatment has an effect	Treatment has no effect
Treatment has an effect	True positive Correct conclusion $1 - \beta$	False positive Type I error α
Treatment has no effect	False negative Type II error β	True negative Correct conclusio $1 - \alpha$

Power of the Test: Chance of detecting a real difference of a given size

Given a method that produces < 0.05 chance of <u>false positive</u> what is the chance of a <u>false negative</u>?



Power of the Test

Figure 6-3 (*A*) The distribution of values of the *t* statistic computed from 200 experiments that consisted of drawing two samples of size 10 each from a single population; this is the distribution we would expect if the diuretic had no effect on urine production. (Compare with Fig. 4-5*A*.) (*B*) The distribution of *t* values from 200 experiments in which the drug increased average urine production by 200 mL/day. $t = 2.1$ defines the most extreme 5 percent of the possible values of *t* when the drug has no effect; 111 of the 200 values of *t* we would expect to observe from our data fall above this point when the drug increases urine production by 200 mL/day. Therefore, there is a 55 percent chance that we will conclude that the drug actually increases urine production from our experiment.

Examples of Errors of Inference

Cardiovascular Disease in Pediatrics

- Suppose average cholesterol level in children is 175 mg/dL
- Cholesterol levels of children of men who died of heart disease in the last year are measured
- Two hypotheses are considered:
  - Average cholesterol level of these children is 175 mg/dL (i.e., not different from general population of children) = null hypothesis (H_0)
  - Average cholesterol level of these children is greater than 175 mg/dL = alternative hypothesis (H_1)

Types of Errors

- Type I error: concluding that offspring of men who died from heart disease have an average cholesterol level greater than 175 mg/dL when in fact their average cholesterol level is 175 mg/dL
- Type II error: concluding that these children have normal cholesterol levels when in fact their cholesterol levels are above average

Possible Consequences

- Type I error: subject children to drugs, diets, exercise regimens, follow-up tests, etc., that are not necessary; waste of money and time, undue worry, drugs and tests might have adverse side effects
- Type II error: do not take preventative action when you should; may result in shorter lifespans and/or negative impacts on quality of life

(Rosner, *Fund of Biostats*)

Examples of Errors of Inference

Birthweight & Socioeconomic Status (SES)

- Birthweights obtained from 100 consecutive, full-term, live-born deliveries from maternity ward of a hospital in a low-SES area
- Mean birthweight found to be 115 oz with standard deviation of 24 oz
- Suppose nationwide mean birthweight (based on millions of deliveries) is 120 oz
- Two hypotheses are considered:
  - Mean birthweight in low-SES area hospital is equal to mean birthweight in U.S. = null hypothesis (H_0)
  - Mean birthweight in low-SES area hospital is less than national average = alternative hypothesis (H_1)

Types of Errors

- Type I error: concluding that mean birthweight in the low-SES area hospital is less than 120 oz when in fact it is 120 oz (i.e., it is the same as the national average)
- Type II error: concluding that mean birthweight at the hospital is 120 oz when in fact it is less than 120 oz

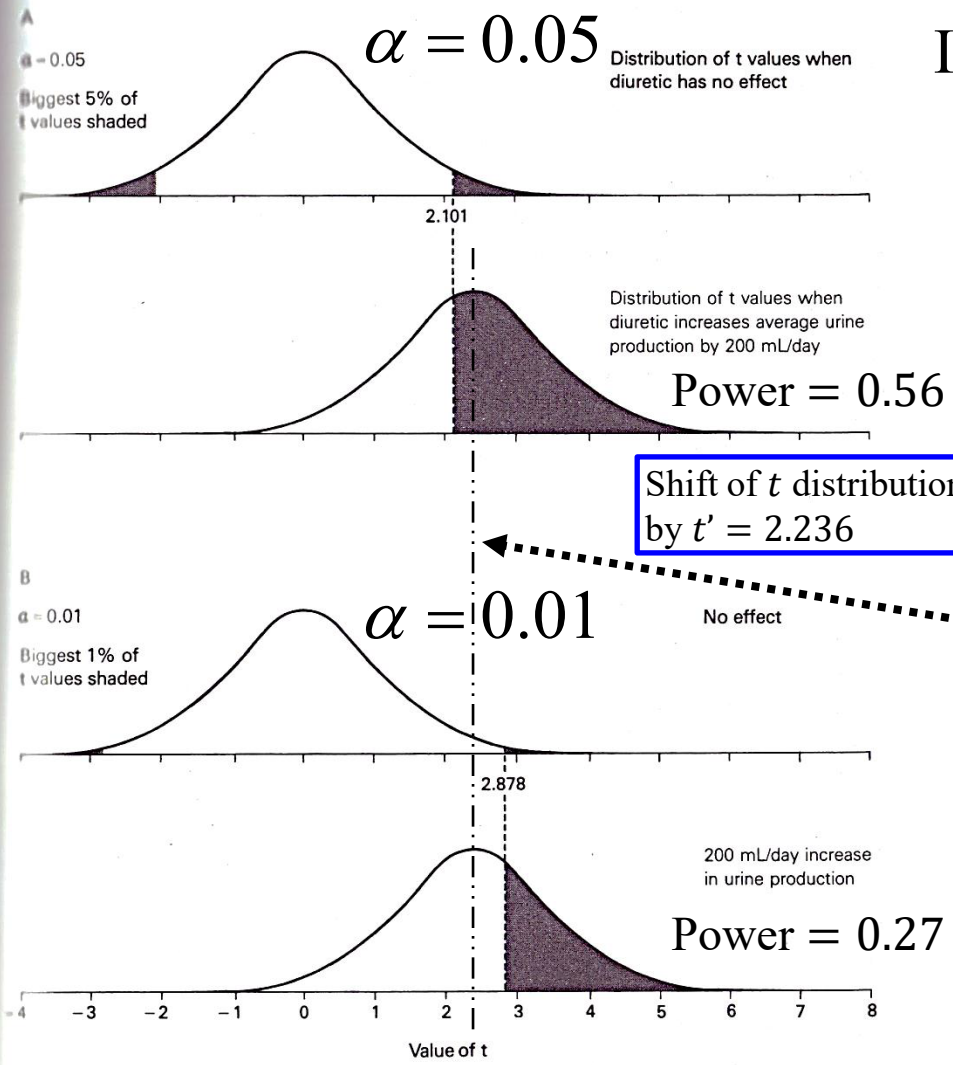
Possible Consequences

- Type I error: an expensive special-care nursery for low-birthweight babies might be built in this hospital when there is actually no need for it
- Type II error: a special-care nursery will not be funded when in fact it is needed; some low-birthweight babies may not survive without the aid of the special-care nursery

(Rosner, *Fund of Biostats*)

III. What Determines a Test's Power

A. Size of Type I Error – α



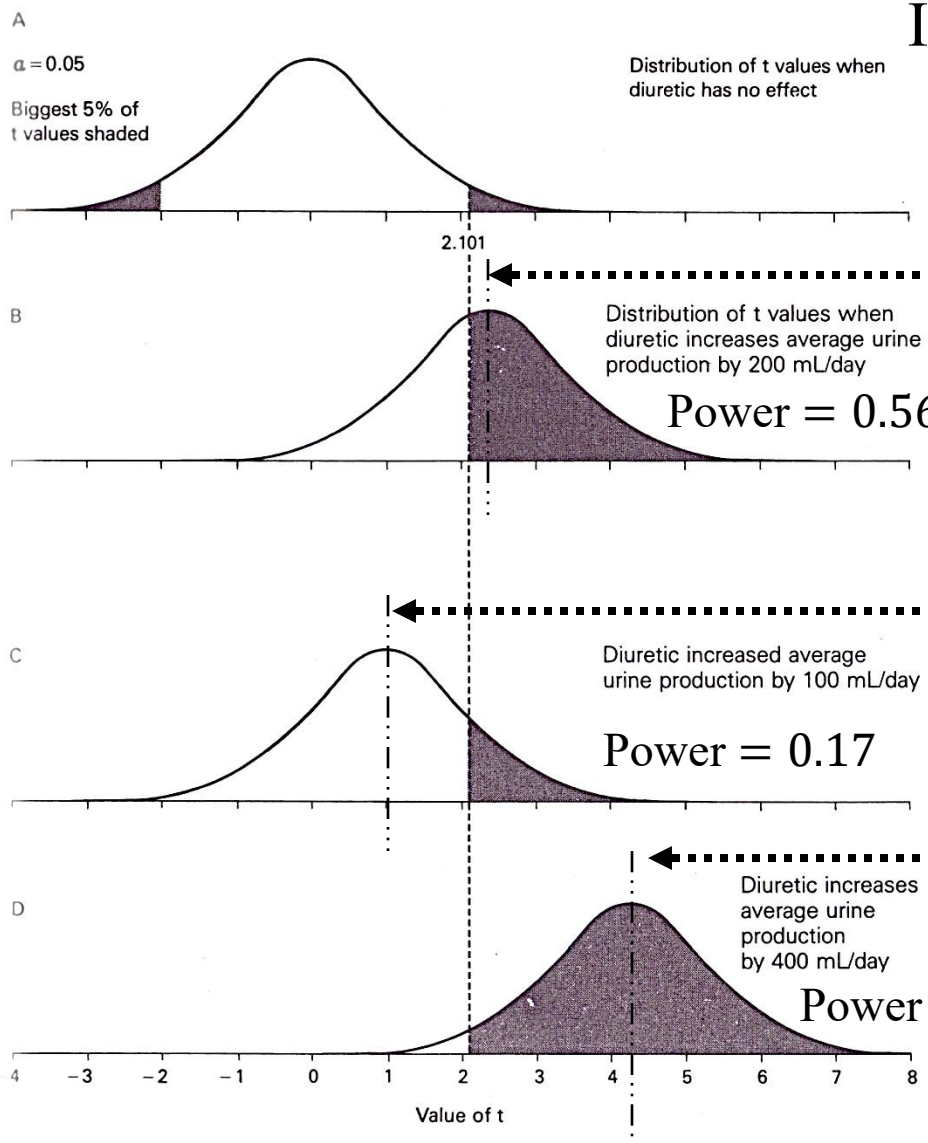
Shift of t distribution by $t' = 2.236$

$$t' = \frac{\mu_1 - \mu_2}{\sqrt{\left(\sigma^2/n_1\right) + \left(\sigma^2/n_2\right)}}$$

$$t' = \frac{200}{\sqrt{\left(200^2/10\right) + \left(200^2/10\right)}} = 2.236$$

Figure 6-4 (*A*) $t = 2.101$ defines the 5 percent most extreme values of the t statistic we would expect to be associated with our experiment if the drug did not increase urine production. Of the possible values of the t statistic that will actually be associated with our experiment, 55 percent fall above this point because the drug increased urine production by 200 mL/day on the average. (*B*) If we increase the confidence with which we wish to reject the hypothesis that the drug had no effect (make α smaller), the proportion of possible t values that will actually arise from the experiment also goes down, so the power of the test to detect an effect of a given size decreases. For example, if we change α from .05 to .01, the power of the test drops from 55 to 45 percent because the cutoff for a "big" t increases from 2.101 to 2.878.

III. What Determines a Test's Power
B. Size of Treatment Effect



$$t' = \frac{200}{\sqrt{\left(200^2/10\right)+\left(200^2/10\right)}} = 2.236$$

$$t' = \frac{100}{\sqrt{\left(200^2/10\right)+\left(200^2/10\right)}} = 1.118$$

$$t' = \frac{400}{\sqrt{\left(200^2/10\right)+\left(200^2/10\right)}} = 4.472$$

Figure 6-5 As the size of the treatment effect increases, the proportion of t values that will actually arise from the experiment increases, so the power of the test increases.

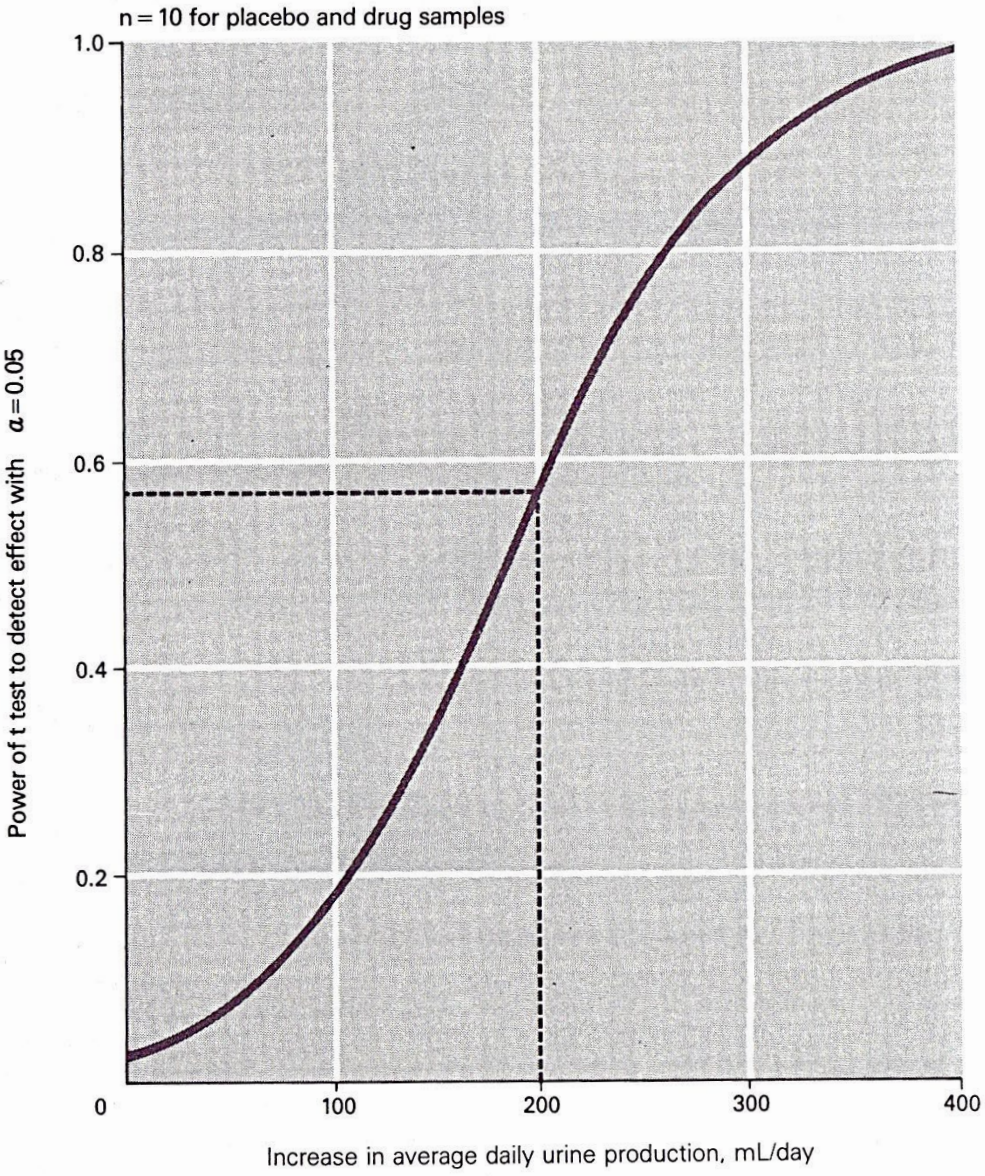


Figure 6-6 The power of a *t* test to detect a change in urine production based on experiments with two groups of people, each containing 10 individuals. The dashed line indicates how to read the graph. A *t* test has a power of .55 for detecting a 200 mL per day change in urine production.

Power Function vs. Size of Treatment Effect

$\alpha = 0.05$
$\sigma = 200$ mL/day
$n = 10$

III. What Determines a Test's Power
C. Population Variability

$$t' = \frac{\mu_1 - \mu_2}{\sqrt{\left(\sigma^2/n_1\right) + \left(\sigma^2/n_2\right)}}$$

$$t' = \frac{\mu_1 - \mu_2}{\sqrt{\left(\sigma^2/n\right) + \left(\sigma^2/n\right)}} = \frac{\mu_1 - \mu_2}{\sigma\sqrt{2/n}}$$

let $\delta = \mu_1 - \mu_2$

$$t' = \frac{\delta/\sigma}{\sqrt{2/n}} = \frac{\delta}{\sigma}\sqrt{\frac{n}{2}}$$

What Determines a Test's Power
D. Sample Size

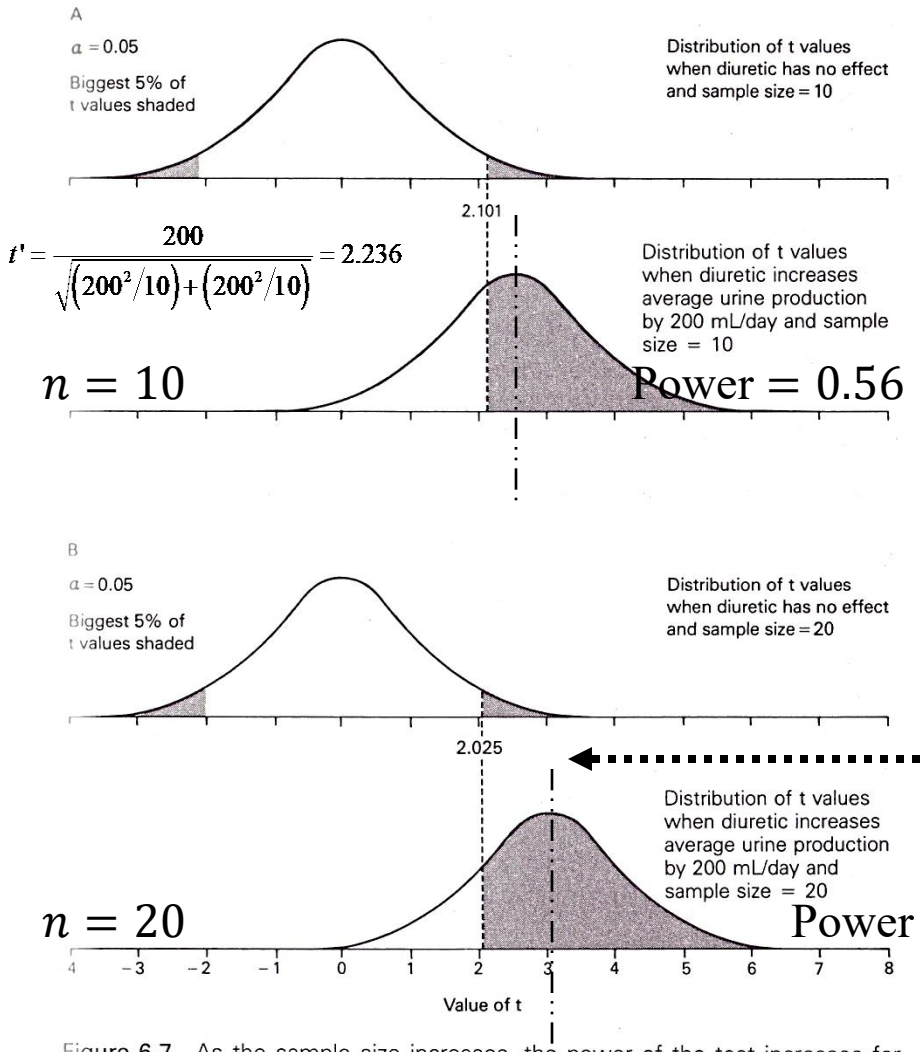


Figure 6-7 As the sample size increases, the power of the test increases for two reasons: (1) the critical value of t for a given confidence level in concluding that the treatment had an effect decreases, and (2) the values of the t statistic associated with the experiment increase.

$$t' = \frac{\mu_1 - \mu_2}{\sqrt{\left(\sigma^2/n_1\right)+\left(\sigma^2/n_2\right)}} = \frac{\delta}{\sigma}\sqrt{\frac{n}{2}}$$

for $n_1 = n_2 = n$

$$t' = \frac{200}{\sqrt{\left(200^2/20\right)+\left(200^2/20\right)}} = 3.162$$

freedom. From Table 4-1 the critical value of t defining the most extreme 5 percent of possible values is 2.025 (compared with 2.101 when there were 10 people in each group). The larger sample size also produces larger values of t, on the average, when the treatment increases urine production by 200 mL/day (with a population standard

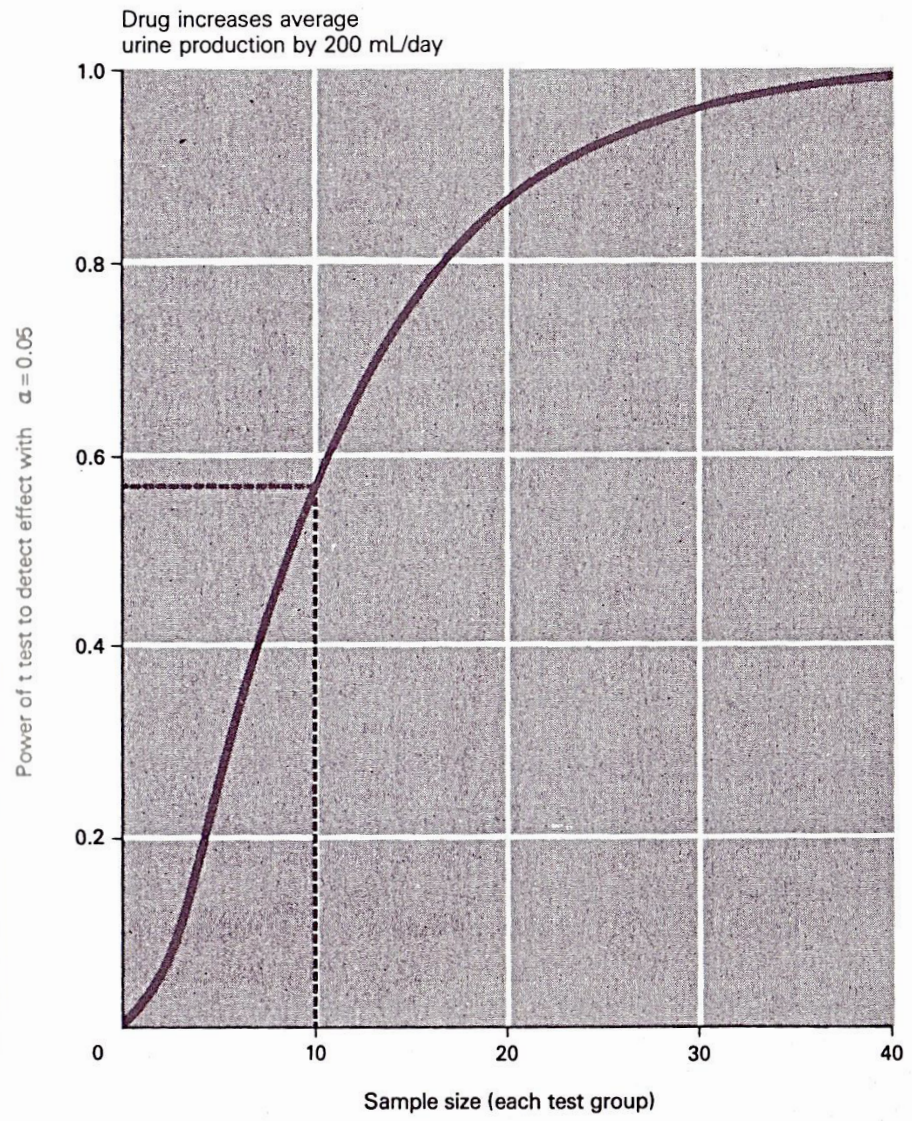


Figure 6-8 The effect of sample size on the power of a *t* test to detect a 200 mL per day increase in urine production with α = .05 and a population standard deviation in urine production of 200 mL per day. The dashed line illustrates how to read the graph. A sample size of 10 yields a power of .55 for a *t* test to detect a 200 mL/day change in urine production.

$\alpha = 0.05$
$\delta = 200$ mL/day
$\sigma = 200$ mL/day

Power Function for t Test

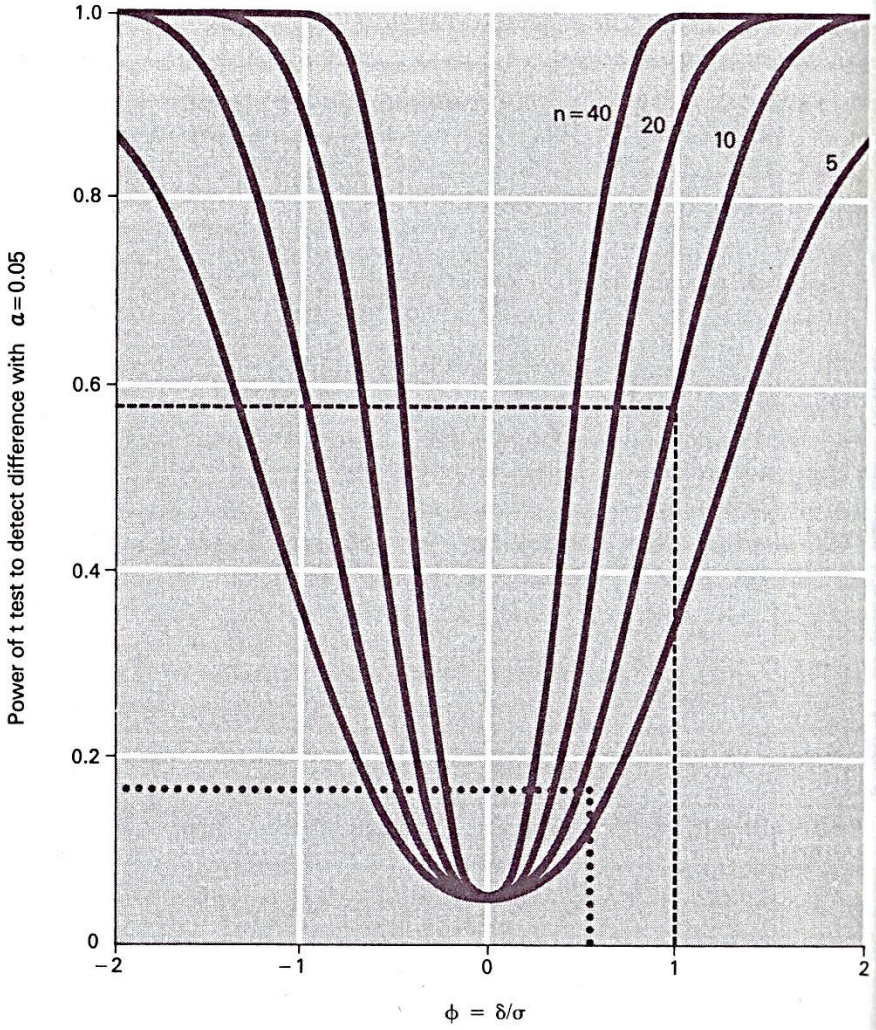


Figure 6-9 The power function for a test for comparing two experimental groups, each of size n, with $\alpha = .05$. δ is the size of the change we wish to detect, σ is the population standard deviation. If we had taken $\alpha = .01$ or any other value, we would have obtained a different set of curves. The dashed line indicates how to read the power of a test to detect a $\delta = 200$ mL/day change in urine production with a $\sigma = 200$ mL/day standard deviation in the underlying population with a sample size of $n = 10$ in each test group; the power of this test is .55. The dotted line indicates how to find the power of an experiment designed to study the effects of anesthesia on the cardiovascular system in which $\phi = \delta/\sigma = .55$ with a sample size of 9; the power of this test is only .16.

$$t' = \frac{\delta/\sigma}{\sqrt{2/n}} = \frac{\delta}{\sigma}\sqrt{\frac{n}{2}}$$

$$\phi = \frac{\delta}{\sigma}$$

$\alpha = 0.05$

IV. Examples – Power & Sample Size for *t* Test

Example 1

Table 4–2 Comparison of Anesthetic Effects on the Cardiovascular System

	Halothane ($n = 9$)		Morphine ($n = 16$)	
	Mean	SD	Mean	SD
Best cardiac index, induction to bypass, $L/m^2 \cdot min$	2.08	1.05	1.75	.88
Mean arterial blood pressure at time of best cardiac index, mmHg	76.8	13.8	91.4	19.6
Total peripheral resistance associated with best cardiac index, $dyn \cdot s/cm^5$	2210	1200	2830	1130

Not Sig; $P > 0.05$

Source: Adapted from T. J. Conahan et al., "A Prospective Random Comparison of Halothane and Morphine for Open-Heart Anesthesia," *Anesthesiology*, **38**:528–535, 1973.

Power & Sample Size for *t* Test - Example

What is the Power to detect a difference in Cardiac Index of $2.08 - 1.75 = 0.33$ L/min/m^2 ?

Pooled Variance

$$s^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2} \qquad n_1 = 9,\ n_2 = 16$$

$$s^2 = 0.89$$

$$\phi \approx \frac{\delta}{s} = \frac{0.33}{\sqrt{0.89}} = \frac{0.33}{0.94} = 0.35$$

Use $n = 9$ (smaller group)

Power ≈ 0.1

10% chance to detect
a difference of this size

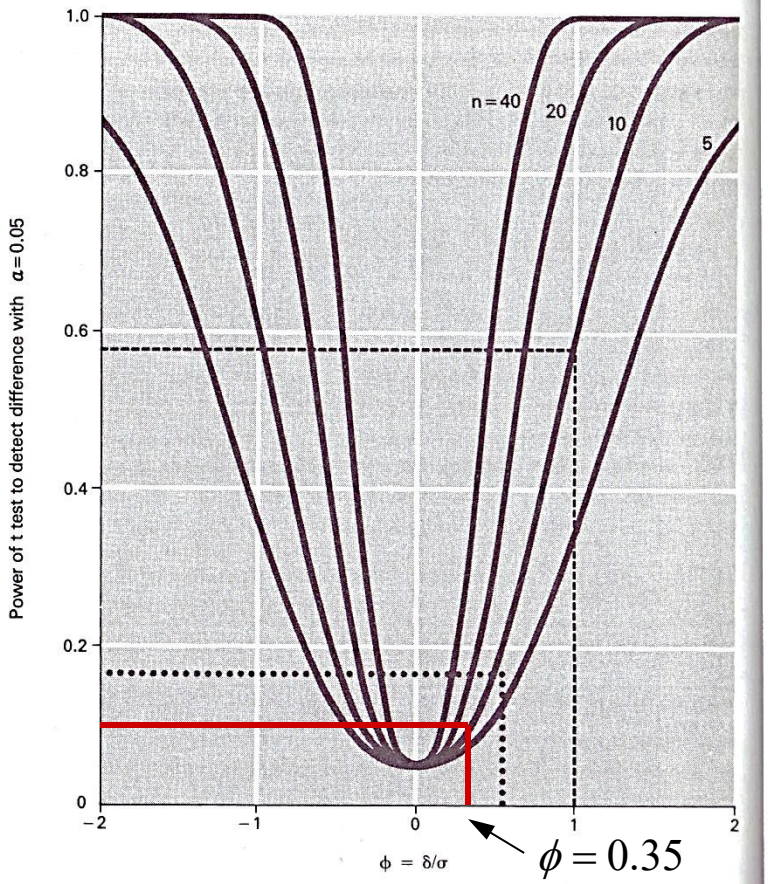


Figure 6-9 The power function for a test for comparing two experimental groups, each of size *n*, with $\alpha = .05$. δ is the size of the change we wish to detect, σ is the population standard deviation. If we had taken $\alpha = .01$ or any other value, we would have obtained a different set of curves. The dashed line indicates how to read the power of a test to detect a $\delta = 200$ mL/day change in urine production with a $\sigma = 200$ mL/day standard deviation in the underlying population with a sample size of $n = 10$ in each test group; the power of this test is .55. The dotted line indicates how to find the power of an experiment designed to study the effects of anesthesia on the cardiovascular system in which $\phi = \delta/\sigma = .55$ with a sample size of 9; the power of this test is only .16.

Power & Sample Size for *t* Test - Example

What is the Sample Size n needed to detect a difference of 0.33 L/min/m^2 with a Power of, say, 0.8 ?

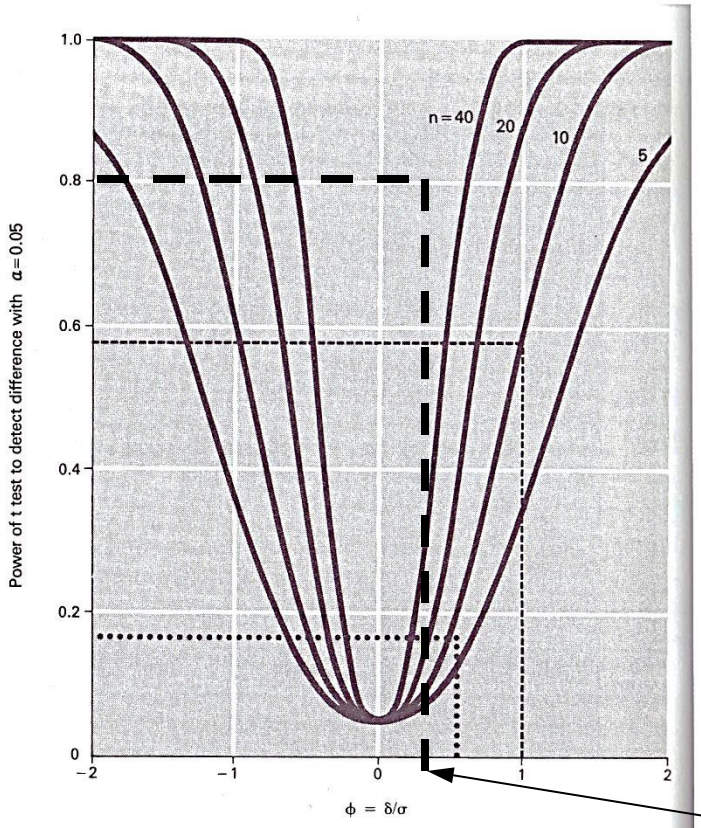


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$$\phi \approx \frac{\delta}{s} = \frac{0.33}{\sqrt{0.89}} = \frac{0.33}{0.94} = 0.35$$

$$n = 129$$

$$\phi = 0.35$$

Power & Sample Size for *t* Test - Example

What is the Sample Size n needed to have an 80% chance of detecting a 25% change in Blood Pressure from that in the Halothane group?

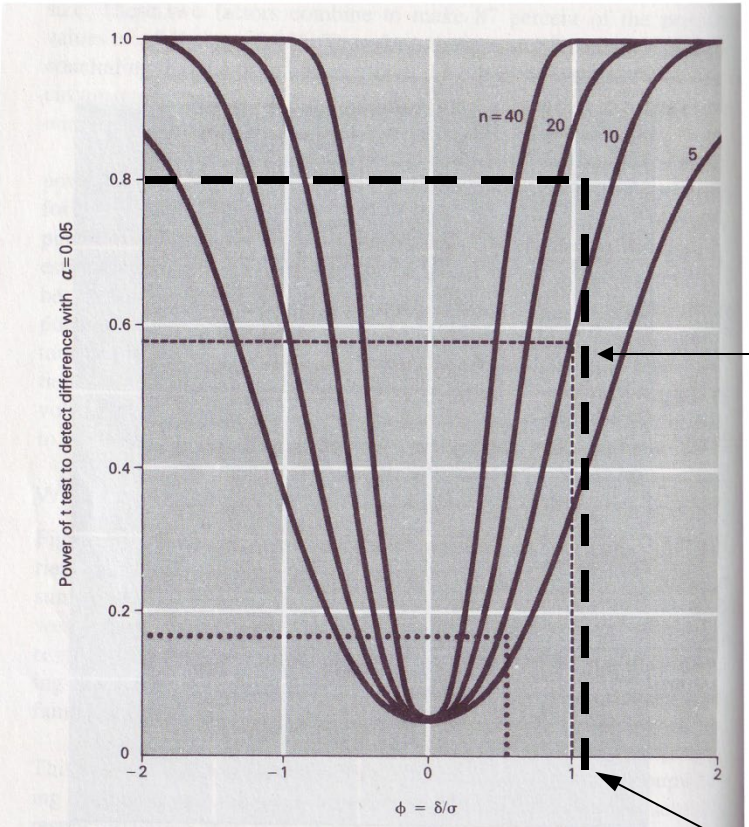


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Pooled Variance $\quad s^2 = 316.8 \quad s = 17.8$

$$\delta = 0.25 * 76.8 = 19.2 \text{ mmHg}$$

$$\phi \approx \frac{\delta}{s} = \frac{19.2}{17.8} = 1.08$$

$\phi = 1.08$

$$n = 15$$

Example 2:

6-1 Both diabetes and high cholesterol interact to increase the risk of heart disease. Changing diet affects both blood sugar and cholesterol. To investigate how different diets used to control diabetes affects cardiovascular risk factors, Neal Barnard and colleagues[†] compared the effects of a low-fat vegan diet with the diet recommended by the American Diabetic Association. What is the power of this study to detect a change in mean total cholesterol from 190 to 165 mg/dL with a sample size of 20 people on each diet with 95% confidence? Based on earlier experience, the standard deviation of total cholesterol in the population is about 35 mg/dL.

$\alpha = 0.05$

$$\phi = \frac{\delta}{\sigma} = \frac{190 - 165}{35} = \frac{25}{35} = 0.71$$

$$n = 20$$

Power Function for t Test

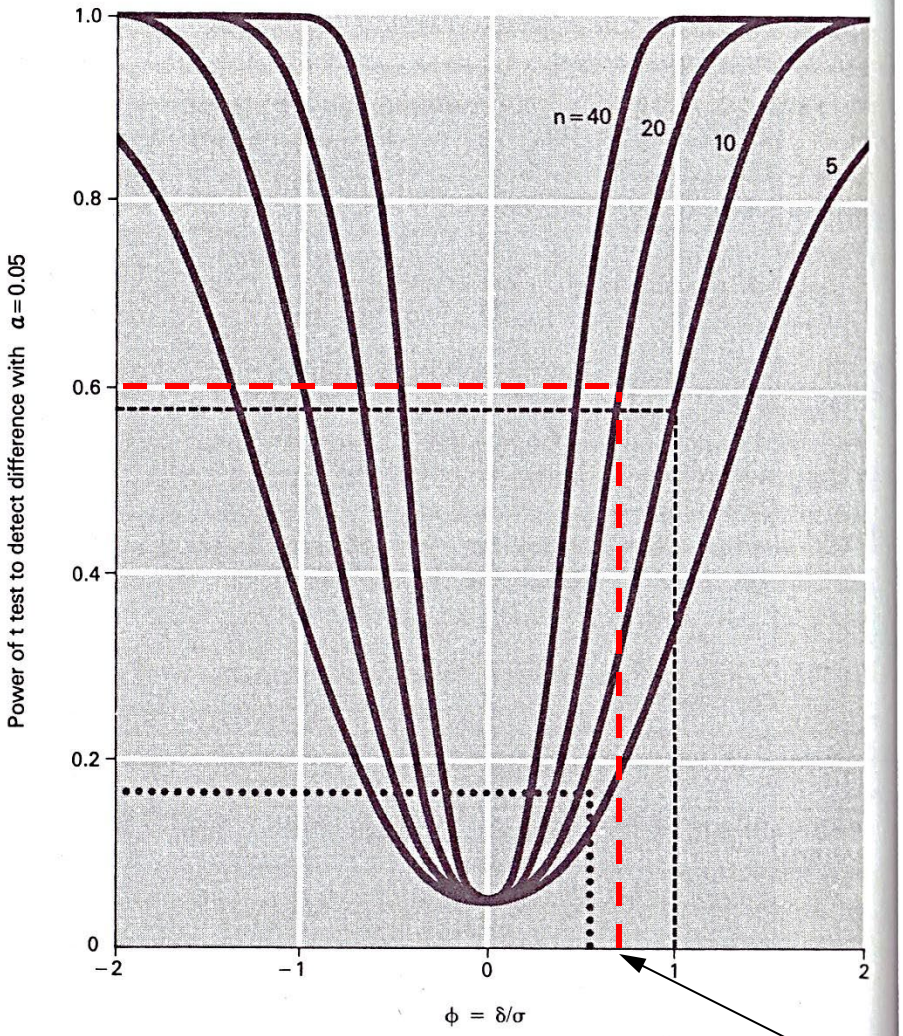


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$\alpha = 0.05$

$\phi = 0.71$

$n = 20$

Power ≈ 0.6

$\phi = 0.71$

6-3 What is the minimum detectable effect one could obtain with 20 people in each group and 80% power?

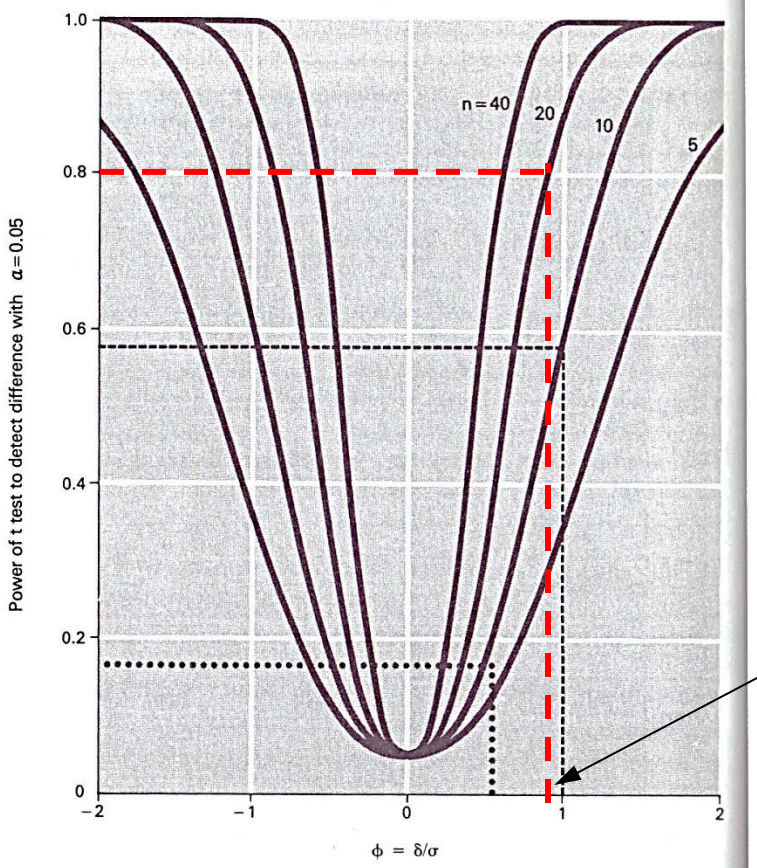


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$\alpha = 0.05$

$n = 20$

Power $= 0.80$

$\phi \approx 0.9$ is needed

$$\phi = \frac{\delta}{\sigma} = \frac{\delta}{35} = 0.9$$

$$\delta = 31.5 \text{ mg/dL}$$

Power of Hypothesis-Testing Procedures – Summary

- Power of a test indicates likelihood that H_0 will be rejected when treatment actually does have an effect
- More stringent level of significance (i.e., smaller chances of a Type I error) results in lower power of the test
- Smaller size of treatment effect relative to the size of the population standard deviation makes treatment effect harder to detect (captured by $\phi = \delta/\sigma$ parameter)
- Larger sample size results in larger power of the test
- With sample data, must approximate true difference of population means and/or true population standard deviation:

$$\phi = \frac{\delta}{\sigma} \approx \frac{\bar{x}_1 - \bar{x}_2}{s}$$

- Exact procedure to compute power of test depends on test itself